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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CASE STUDY 2

TOPIC: Urban Air Quality Analysis

P24CS904

DATA SCIENCE

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1.Abstract

Air pollution is a major environmental issue in Indian cities. This project analyses the air quality data of Indian cities from 2015 to 2020 using Python. It focuses on PM2.5 and NO_2 pollution levels, seasonal variations and city-wise comparisons. Statistical methods such as confidence intervals, t-tests and Mann–Whitney U tests are used to compare pollution levels during the 2019 and 2020 COVID-19 lockdown periods. Correlation analysis is also performed to understand relationships between different pollutants. The project aims to identify pollution patterns and provide insights that support environmental awareness and sustainable urban development.

2. Introduction

Air pollution is a major environmental problem in Indian cities. Pollutants such as PM2.5 and NO_2 affect air quality and human health.

During the COVID-19 lockdown in 2020, transportation and many industrial activities were restricted. This project investigates whether air pollution levels changed during this period compared with 2019.

3. Objectives

1. Analyse PM2.5 and NO_2 levels in Indian cities.
2. Compare pollution levels across different seasons.
3. Identify cities with higher pollution levels.
4. Compare 2019 and 2020 lockdown-period pollution.
5. Calculate confidence intervals.
6. Study relationships between pollutants.
7. Explain why correlation does not mean causation.

4. Dataset Description

Dataset: CPCB City-Day Air Quality Data, India (2015–2020).

Source: Air Quality Data in India .

Column	Description
City	Indian city name
Date	Measurement date
PM2.5	Fine particulate matter
PM10	Particulate matter

NO2	Nitrogen dioxide
SO2	Sulphur dioxide
CO	Carbon monoxide
O3	Ozone
AQI	Air Quality Index

5. Tools and Technologies

- Python
- Pandas
- NumPy
- SciPy
- Statsmodels
- Matplotlib
- Seaborn
- Jupyter Notebook

6. Data Cleaning

Simple cleaning steps:

- Load the dataset.
- Check missing values.
- Remove duplicate records.
- Convert Date into datetime format.
- Handle missing PM2.5 and NO_2 values.
- Check extreme pollution values.
- Create Year, Month and Season columns.

Python code:

python

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("city_day.csv")

df["Date"] = pd.to_datetime(df["Date"])

df = df.drop_duplicates()

print(df.head())
print(df.isnull().sum())
```

7. Exploratory Data Analysis (EDA)

Topic 1: City-wise PM2.5 analysis

Find the average PM2.5 concentration in each city.

python

```
city_pm = df.groupby("City")["PM2.5"].mean()

city_pm.sort_values().plot(
   kind="bar",
   figsize=(12,5)
)

plt.title("Average PM2.5 by City")
plt.show()
```

Topic 2: Seasonal pollution analysis

Create four seasons:

- Winter
- Summer
- Monsoon
- Post-Monsoon

python

```
df["Month"] = df["Date"].dt.month

def season(m):
   if m in [12, 1, 2]:
      return "Winter"
```

```
    elif m in [3, 4, 5]:
        return "Summer"
    elif m in [6, 7, 8, 9]:
        return "Monsoon"
    else:
        return "Post-Monsoon"

df["Season"] = df["Month"].apply(season)

sns.boxplot(
    data=df,
    x="Season",
    y="PM2.5"
)

plt.show()
```

Topic 3: NO_2 analysis

Study how nitrogen dioxide levels vary across cities.

python

```
city_no2 = df.groupby("City")["NO2"].mean()

city_no2.plot(
    kind="bar",
    figsize=(12,5),
    color="orange"
)

plt.title("Average NO2 by City")
plt.show()
```

8. Confidence Interval Analysis

A confidence interval estimates a range of plausible values for the population mean.

For this project, use a 95% confidence interval for average PM2.5.

python

```
from scipy import stats

data = df["PM2.5"].dropna()

mean = data.mean()
sem = stats.sem(data)

ci = stats.t.interval(
    0.95,
    len(data)-1,
    loc=mean,
    scale=sem
)

print("Mean:", mean)
print("95% Confidence Interval:", ci)
```

Interpretation: The confidence interval provides an estimate of uncertainty around the sample mean.

9. Hypothesis Testing: 2019 vs 2020 Lockdown

Research question: Did PM2.5 pollution change during the 2020 lockdown compared with the corresponding period in 2019?

For a simple comparison, use 25 March–31 May in both years.

- H0: There is no difference in PM2.5 levels between the two periods.
- H1: There is a difference in PM2.5 levels between the two periods.

Python code – Mann-Whitney U test

python

```
from scipy.stats import mannwhitneyu

df["Year"] = df["Date"].dt.year

df["Month"] = df["Date"].dt.month

period_2019 = df[
    (df["Year"] == 2019) &
```

```
    (
        (df["Date"].dt.month > 3) |
        (
            (df["Date"].dt.month == 3) &
            (df["Date"].dt.day >= 25)
        )
    ) &
    (
        (df["Date"].dt.month < 6)
    )
]["PM2.5"].dropna()

period_2020 = df[
    (df["Year"] == 2020) &
    (
        (df["Date"].dt.month > 3) |
        (
            (df["Date"].dt.month == 3) &
            (df["Date"].dt.day >= 25)
        )
    ) &
    (
        (df["Date"].dt.month < 6)
    )
]["PM2.5"].dropna()

stat, p = mannwhitneyu(
    period_2019,
    period_2020,
    alternative="two-sided"
)

print("P-value:", p)
```

Decision:

- If p-value is below 0.05: Reject H0.
- If p-value is 0.05 or above: Fail to reject H0.

For a t-test, check its assumptions and use a Welch independent-samples t-test where appropriate.

For stronger comparisons, match the same cities and calendar dates, and account for repeated daily observations and seasonal patterns.

10. Correlation Analysis

Correlation measures the relationship between two variables.

Example: PM2.5 and PM10.

python

```
corr = df[
   ["PM2.5", "PM10", "NO2", "SO2", "CO", "O3"]
].corr()


sns.heatmap(
   corr,
   annot=True,
   cmap="coolwarm"
)


plt.title("Pollutant Correlation")
plt.show()
```

Learn more

Why correlation does not mean causation

If PM2.5 and NO_2 increase together, it does not automatically mean one pollutant causes the other.

Both may be influenced by:

- Vehicle traffic.
- Industrial activities.
- Weather conditions.
- Temperature.
- Wind speed.
- Seasonal changes.

Correlation only identifies an association, not a direct cause-and-effect relationship.

11. Expected Findings and Conclusion

The completed analysis should answer these questions:

1. Which cities have higher average PM2.5 levels?
2. Which season experiences higher pollution?
3. Did PM2.5 levels differ between the 2019 and 2020 lockdown-comparison periods?
4. Are PM2.5 and NO_2 correlated?
5. What environmental factors might explain the observed patterns?

12.Conclusion

This project uses Python-based data analysis to understand air pollution in Indian cities.

It combines data cleaning, exploratory data analysis, confidence intervals, hypothesis testing and correlation analysis.

The comparison of 2019 and 2020 provides an opportunity to investigate changes in pollution during the COVID-19 lockdown period.

The findings can support environmental awareness and sustainable urban planning.

13. References

1. CPCB – Central Pollution Control Board: https://cpcb.nic.in/
2. Kaggle – Air Quality Data in India: https://www.kaggle.com/datasets/rohanrao/air-quality-data-in-india
3. SciPy Documentation: https://docs.scipy.org/doc/scipy/
4. Statsmodels Documentation: https://www.statsmodels.org/
5. WHO Air Quality Guidelines: https://www.who.int/health-topics/air-pollution